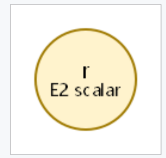
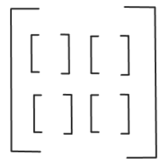
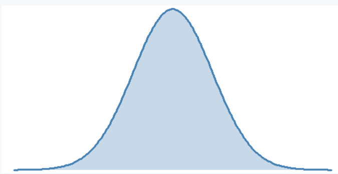
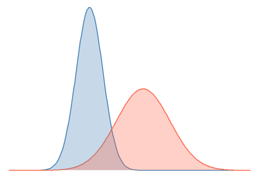
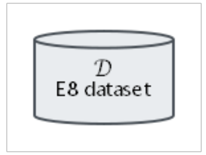
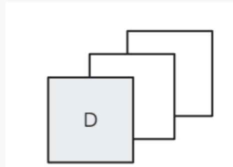
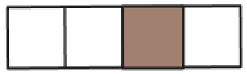
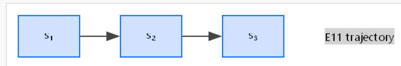
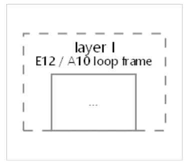
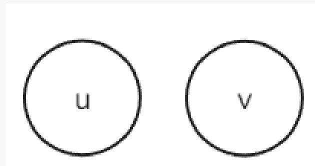
PaperLab visualizer dictionary

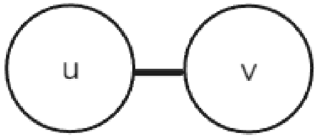
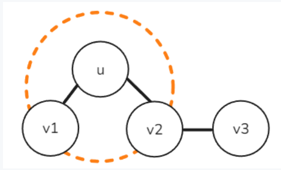
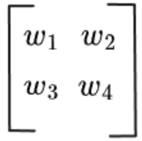
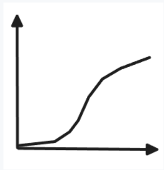
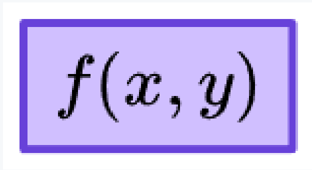
Auto-generated from .cursor/skills/ml-visualization/DICTIONARY.md. Do not edit this PDF by hand — edit the markdown source and rerun python -m tools.build_dictionary_pdf. To attach a hand-drawn symbol to a row, save a PNG under .cursor/skills/ml-visualization/symbols/ and add an inline `![](symbols/<id>.png)` to that row's Symbolic-representation cell.

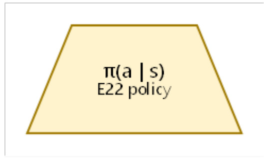
72 entries total · **31** with user-drawn symbol · **18** with graphviz auto-render · **23** placeholder.

Entities (23)

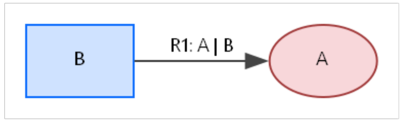
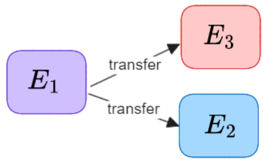
#	Canonical name	Aliases	Symbolic representation	Symbol
E1	vector	latent code, embedding, hidden state, representation z, h, s, Z_X	a vertical column of small cells (cell-column chip); cell count ~ dimensionality	
E2	scalar	reward, probability, log-likelihood, weight ρ, threshold, hyperparameter	a small filled disc, or a number printed inline; size ~ magnitude when meaningful	
E3	tensor / image	observation o_t, frame x_t, RGB image	a square or rectangle with optional grid; for images, a thumbnail; spatial axes preserved	
E4	distribution	prior p(z), posterior p(z| x), likelihood p(x| z), proposal q(x), marginal p(y)	a smooth density curve (continuous) or a histogram / bar set (discrete); the *shape* of the curve carries the distribution's character	
E5	conditional distribution	p_θ(s_t|…), q_ϕ(z| x), p(z_t+1| a_t, z_t, h_t)	a distribution shape (as E4) sitting inside a node, with incoming arrows from the conditioning variables	

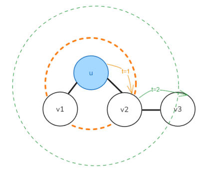
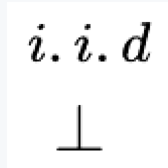
#	Canonical name	Aliases	Symbolic representation	Symbol
E6	sufficient statistics (μ, σ)	encoder outputs, Gaussian parameters (μ, σ^2)	a vector chip (as E1) split into two named halves; top half labeled μ, bottom half labeled σ or σ^2	$\bar{X}; \sigma$
E7	parameter set	θ, ϕ, ψ, ρ (learnable)	a small square with a hat or color code; attached to the function it parameterizes via a thin line (not an arrow)	$\{\theta, \phi, \gamma, \rho, \lambda\}$
E8	dataset	\mathcal D, \mathcal X = \x^(i)\, rollout buffer	a stack of cards or a labeled bin; size indicates "large collection"	
E9	minibatch	X_M, B sequences, K-sample batch	a small cluster of sample markers, drawn from the dataset bin via a sampling arrow	
E10	sample (a single realization)	x^(i), ε^(l), z^(i,l), x_i ~ q	a single marker — a dot, a tagged cell, or a highlighted node	
E11	trajectory / sequence	\(s_τ, a_τ)_τ=t^t+H, \x_t_t=1^L, time series	a horizontal chain of entities (one per step) connected by next-step arrows	
E12	loop index / operational substrate	layer l, time t, hop t, epoch, generation, IS sample i	a labeled frame (rectangle or band) enclosing the per-iteration content, with the loop name written on the frame	
E13	graph node	v, u, vertex	a filled circle, optionally with a name inside	

#	Canonical name	Aliases	Symbolic representation	Symbol
E14	graph edge / adjacency	A, E, edge (u,v)	a line connecting two nodes; arrow if directed	
E15	candidate set / neighborhood	hop pool V_vt, k-NN candidates, leave-one-out negatives	a dashed ring (or enclosure) around an anchor entity, containing the candidate nodes on its boundary	
E16	learnable weight matrix	W^(l), W_c, W_out, W_2 W_1	a small grid (rows × cols); attached to a function as its parameter (see E7)	
E17	nonlinearity	tanh, ReLU, LeakyReLU, σ	a small operator glyph, e.g., a "kink" curve for ReLU or "S" for sigmoid; placed inline on an arrow	
E18	noise variable	ε ~ \mathcal N(0, I), Gumbel noise	a small jittered marker or a stylized "ε" disc; usually injected into a reparameterization step	
E19	critic / energy function	f(x, y)	a labeled black-box rectangle that takes two inputs and emits a scalar	
E20	recurrent state	h_t (LSTM hidden), c_t (LSTM cell)	a vector chip (as E1) with a self-loop arrow back to itself (the recurrence)	*— no tile —*
E21	reference distribution	p_θ(z) = \mathcal N(0, I), Q(Z_A) uniform / Bern(α), q(y) as variational marginal	same shape as a distribution (E4), rendered in grey or with a "ghost" outline; signals "this is what we're comparing to"	*— no tile —*

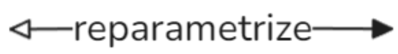
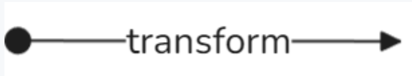
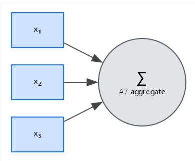
#	Canonical name	Aliases	Symbolic representation	Symbol
E22	controller / policy	$q_\phi(a \mid s)$, $a = W_c[z; h]$, π	a labeled trapezoid or rectangle taking state in, emitting action out	
E23	terminal event / done flag	d_t, episode end, $done_t$	a small "stop" glyph (filled square or X) attached at the time-step where termination is decided	

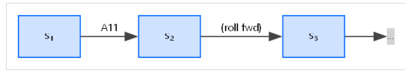
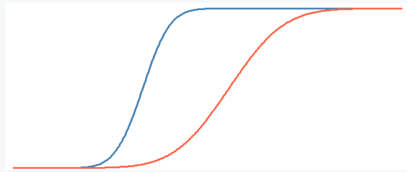
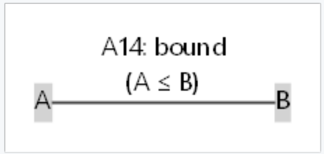
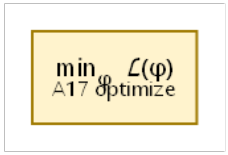
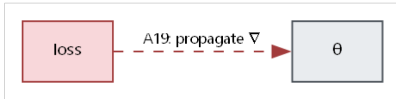
Relations (12)

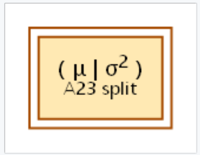
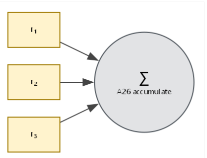
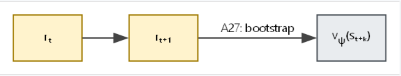
#	Canonical name	Aliases	Symbolic representation	Symbol
R1	conditional dependence	"A | B", "given B", p(A | B)	directed arrow from the condition B into the outcome A	
R2	temporal next-step	A_t → A_t+1, "next step", "evolves to"	horizontal arrow from A_t to A_t+1, optionally annotated with the transition operator	
R3	parameterized-by	f_θ, p_θ, W_c	parameter symbol attached to the function via a thin line (not an arrow); the function and parameter sit close together	
R4	membership	u ∈ S, x_i ∈ X_M, T ∈ \mathcal T	the member sits on (or inside) the boundary of the containing enclosure	*— no tile —*
R5	expectation scope	■_p(x)[·], "average under p"	a containing brace or shaded background labeled with the distribution; the averaged expression sits inside	*— no tile —*
R6	bound / ordering	A ≤ B, "upper-bounds", "lower-bounds", "tightens to"	write the math symbol (≤, ≥, =, ≈) directly; no further visual idiom needed (see Conventions)	$A < x < B$ $[A, B]$ A············x············B
R7	decomposition	A = B_1 + B_2 + …, "A splits into"	a large brace or parenthesis labeled A on the outside, containing B_1, B_2, … inside; each part labeled	$A \begin{Bmatrix} B_1 \\ B_2 \\ \dots \\ B_n \end{Bmatrix}$
R8	stop-gradient barrier	sg(·), "frozen during", "target network"	a perpendicular bar across an arrow (like a one-way valve); gradient cannot flow back through it	
R9	transfer / reuse	"trained in E_1, deployed in E_2", "policy transfer"	the same module drawn once with two arrows pointing into two different context boxes	

#	Canonical name	Aliases	Symbolic representation	Symbol
R10	hop / graph distance	d(u, v) = t	concentric rings around an anchor node, one ring per hop t; candidate nodes sit on the ring matching their hop	
R11	i.i.d.	"i.i.d.", "x_i ~ p independently"	multiple sample markers drawn identically, with a small "i.i.d." badge or repeated arrows from the same distribution	
R12	composition (chained modules)	f = f_2 ■ f_1, "V → M → C", "encoder → sample → decoder"	modules drawn as boxes in a horizontal or vertical chain, connected by single arrows; the chain itself is the relation	

Actions (37)

#	Canonical name	Aliases	Symbolic representation	Symbol
A1	sample	"~", samples, draws, drawn iid from	arrow from a distribution shape (E4) to a sample marker (E10); optionally with a small die or dice glyph at the action point	
A2	draw	"draw from", "sample n from", "drawn iid from"	arrow from a dataset bin (E8) or proposal (E4) to a minibatch cluster (E9); the cluster is the action's output	
A3	encode	"embed", "recognize"	arrow from a data entity (E3) into a labeled encoder module, emerging as a latent vector (E1)	
A4	decode	"reconstruct", "generate"	arrow from a latent vector (E1) into a labeled decoder module, emerging as a data entity (E3)	
A5	reparameterize	"Gumbel-softmax", "reparameterization trick"	one arrow from sufficient statistics (E6) into the sample (E10); the noise variable (E18) feeds the *same* arrow's head from the side (must be drawn adjacent to A5's head, not floated far away). The sample is differentiable along the (μ, σ) path	
A6	transform	"project", "linear-map", "apply W"	arrow from input vector through a labeled box (the weight matrix E16) to output vector	
A7	aggregate	"pool", "sum-pool", "sum over", "mean", "mixture sum"	multiple arrows converging into a single Σ glyph (or similar aggregator), with one outgoing arrow to the aggregated result. Specializations: partition function Z = Σ_x f(x) is "aggregate over all configurations of x"	
A8	attend	"softmax over", "score each u", "attention"	rays from a query entity to each member of a candidate set (E15), with ray width or brightness proportional to the attention weight; emits a distribution (E4) over the set	

#	Canonical name	Aliases	Symbolic representation	Symbol
A9	predict	"infer", "classify", "output"	arrow from a vector through a labeled prediction head into a labeled output (class label, scalar, etc.)	
A10	iterate	"loop", "for", "while"	a labeled frame around the per-iteration content (uses E12 loop-index entity); the loop variable named on the frame	*— no tile —*
A11	roll forward	"imagine", "dream", "rollout", "sample sequentially", "unroll"	a chain of repeated single-step samples (A1) drawn end-to-end as a horizontal sequence of arrows, each step's output feeding the next step's input	
A12	compare	"KL", "divergence from", "distance from"	two distribution shapes (E4 and E21) side by side, with a labeled gap or double-arrow between them; the gap *is* the comparison's magnitude	
A13	measure	"mutual information", "I(X;Y)" — object is "dependency"	two random variables with a labeled overlap glyph (info-diagram lens) or a directed double-arrow labeled I; sometimes drawn as a Venn-style intersection	*— no tile —*
A14	bound	"upper-bounds", "lower-bounds", "prove A ≤ B"	the bounded quantity drawn with a horizontal line above (upper) or below (lower) it; relation written with the math symbol ≤ / ≥ (see Conventions)	
A15	sandwich	" Z^- ≤ Z ≤ Z^+", "combine upper and lower"	the target quantity drawn between two horizontal lines (upper bound above, lower bound below); the vertical interval is the sandwich	*— no tile —*
A16	tighten	"tightens as n → ∞", "any-time refinement"	a bound line (from A14) moving closer to the bounded quantity over an arrow labeled with the iteration / sample count	*— no tile —*
A17	optimize	"maximize", "minimize", max_ϕ, min_ψ, SGD, Adam, CMA-ES	a labeled max or min glyph attached to the objective; arrow indicating direction of improvement (up for max, down for min)	
A18	update	"gradient step", "Adam tick", "θ arrow θ + η g"	an arrow from the gradient into the parameter (E7), often labeled with the optimizer step rule	*— no tile —*
A19	propagate	"backprop through", "analytic gradients through" — object is "gradients"	a dashed reverse arrow drawn alongside the forward dataflow, going right-to-left (or backward along the chain)	

#	Canonical name	Aliases	Symbolic representation	Symbol
A20	freeze	"frozen during", "held fixed"	a snowflake or padlock glyph attached to the parameter (E7); during the freeze, no gradient arrow reaches it	
A21	compose	"f_2 ■ f_1", "chain", "stack layers"	the chained modules drawn as boxes connected by single arrows; the composition arises visually from the chain	*— no tile —*
A22	concatenate	"stack", "[z; h]", "⊕"	two vector chips (E1) drawn end-to-end, with a small bracket or ";" between them; the joined chip is the output	*— no tile —*
A23	split	"first / second half", "partition into"	one vector chip (E1) with a visible mid-cut; the two halves labeled separately (e.g., as μ and σ)	
A24	squash	tanh, sigmoid, "bound to range"	a small "S"-curve glyph inline on the arrow; the output range labeled	*— no tile —*
A25	execute	"act in env", "env.step"	arrow from action (E22 output) into an "env" box; environment returns observation and reward on the outgoing arrow	*— no tile —*
A26	accumulate	"Σ_t r_t", "running sum", "fitness = mean return"	scalar samples flowing into a Σ glyph along a sequence (E11); emits a single aggregated scalar	
A27	bootstrap	"n-step bootstrap", "value tail"	a partial rollout (chain of arrows) terminated by a labeled value glyph v(·); the value glyph supplies the "tail"	
A28	interpolate	"blend", "mix between"	two endpoint shapes drawn with a parameter-controlled axis between them; intermediate shapes shown along the axis	*— no tile —*
A29	construct	"form", "build" — object is a distribution or proposal	inputs (components, scores) flowing into a labeled "construct" glyph, producing a distribution shape (E4)	*— no tile —*
A30	estimate	"MC mean", "sample mean", " Z = \frac1nΣ w_i"	n sample markers flowing into a \frac1nΣ glyph, emitting a single estimated scalar	*— no tile —*
A31	evaluate	"analytic", "closed-form" — object is a deterministic function value	inputs flowing into a labeled compute-box (no random nodes inside), emitting the exact scalar	*— no tile —*
A32	threshold	"kept if", "Bern keep/drop", "done > 0.5"	a horizontal cutoff line on a scalar axis; outcomes above and below labeled (kept / dropped, true / false)	*— no tile —*

#	Canonical name	Aliases	Symbolic representation	Symbol
A33	constrain	"subject to", "Σ w = 1", "TC ≤ δ"	a lock or clip glyph applied to the constrained quantity, with the constraint formula written adjacent in math (see Conventions)	*— no tile —*
A34	regularize	"penalize", "+ β · KL"	an additive term drawn next to the main objective with a coefficient (β) glyph; the penalty's source (a comparison or distance) shown attached	*— no tile —*
A35	perturb	"add noise", "exploration noise"	a small jitter glyph injected onto an arrow; the noise distribution labeled at the injection point	*— no tile —*
A36	approximate	"q ≈ p", "variational approximation"	the approximating distribution drawn solid; the target drawn ghostly behind it; an "≈" glyph between them	*— no tile —*
A37	divide	"ratio", "w = f/q", "density ratio"	two scalars (E2) feeding into a fraction bar (numerator over denominator), emitting a scalar	*— no tile —*